Danmarks Tekniske Universitet

42879 - Decision Support and Strategic Assessment

Group 49 - Project Assignment

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1 Introduction

Group 49 has received data with regard to 10 infrastructure scenarios belonging to EU's Trans-European Network (TEN) project portfolio. The objective of the report is to find the most attractive scenario from an economic and strategic perspective, ultimately concluding the results with a recommendation for the EU Commission. Table 21 in the Appendix contains the scenarios and their corresponding TEN-T priority projects, additional information and details used for the initial screening analysis with regard to the priority projects are found on the European Commission's INEA website [1].

The appraisal study is based on the following workflow:

- **Screening Analysis:** Define the selection criteria and conduct a screening analysis using the ELECTRE method, reducing the 10 scenarios to 4. with the most strategic impact.
- **Socioeconomic Analysis:** Conduct a Cost-Benefit Analysis (CBA) on the 4 scenarios using the given impact and consumer surplus data provided.
- **Feasibility Risk Assessment (FRA):** Conduct an FRA on the 4 scenarios using Monte-Carlo simulations of 2000 samples for each scenario.
- **Strategic Analysis:** Perform a multi-criteria decision analysis (MCDA) for the 4 scenarios using select MCDA techniques, identifying at least five strategic impacts complementing the CBA.
- **Composite Assessment:** Using a composite analysis, the CBA and MCDA are combined into one comprehensive assessment depicting the mist attractive scenario for the EU Commission to consider.

2 Selection and Definition of Criteria

2.1 Environmental impact (weight: 4)

Environmental impact is a central consideration in European Union transport planning because projects must comply with several legally binding frameworks, including the Habitats Directive, Birds Directive, and Water Framework Directive, as well as strategic documents such as the European Green Deal and the European Union Biodiversity Strategy for 2030. Transport infrastructure can significantly affect terrestrial, aquatic, and marine ecosystems through habitat fragmentation, soil and groundwater disruption, changes in water flow, species disturbance, and long-term landscape modification. Construction also has a measurable carbon footprint, especially for projects requiring tunneling, large quantities of cement and steel, or major earthworks. Furthermore, the European Union increasingly evaluates transport investments in terms of their influence on fossil fuel dependency. Projects that facilitate a shift from road to rail or waterborne transport contribute positively to climate goals, while those that risk inducing additional road traffic may increase overall emissions.

Reason for weight 4: Environmental performance is important and regulated, but within the Trans European Transport Network framework it balances against broader connectivity and economic objectives. A mid-level weight reflects its significance without allowing it to dominate the evaluation.
Definition for ELECTRE: The degree to which the project causes direct or indirect harm to ecosystems, biodiversity, water systems, landscape integrity, and climate objectives, including the project's carbon footprint and long-term fossil fuel implications.

2.2 Cross-border connectivity and network integration (weight: 8)

Cross-border connectivity is one of the principal goals of the Trans European Transport Network policy. Many inefficiencies in European freight and passenger transport occur at borders where infrastructure, regulations, and technical standards do not align. Improving these connections reduces travel times, increases reliability, eliminates bottlenecks, and strengthens the functioning of the European Single Market. Network integration also includes harmonizing technical systems such as electrification standards, axle-load capacities, track gauge compatibility, and deployment of the European Rail Traffic Management System. A well-integrated network ensures that investments in one segment generate benefits across multiple member states. Because the European Union treats transport as a shared strategic asset, resolving cross-border gaps yields especially high added value.
Reason for weight 8: This criterion has very high relevance because it determines whether a project delivers European-level benefits rather than merely national improvements. Its weight reflects the central role of integration in Trans European Transport Network policy.
Definition for ELECTRE: The extent to which the project removes cross-border bottlenecks, improves interoperability, supports harmonized technical standards, and enhances integration within the core European network.

2.3 Cost (weight: 6)

Cost is a fundamental parameter in infrastructure prioritization because the European Union and its member states operate within limited investment budgets. High-cost projects can strain national public finances, reduce the availability of Connecting Europe Facility funding for other corridors, and introduce long-term fiscal commitments. In this context, the cost criterion refers to the absolute capital expenditure required to build the project, independent of later cost-benefit analyses. Large-scale investments also have higher exposure to risks such as inflation, material shortages, workforce constraints, and unexpected geological or engineering challenges. Evaluating total cost at the prioritization stage ensures that decisions remain financially realistic and that the overall European transport strategy can be implemented in a balanced and sustainable way.
Reason for weight 6: Cost must carry significant influence because financial feasibility constrains what can be delivered. However, it is secondary to socio-economic impact and integration, so it is assigned a moderately high but not dominant weight.
Definition for ELECTRE: The overall financial magnitude of the project, measured as total capital expenditure and required public investment.

2.4 Complexity and feasibility (weight: 4)

Complexity and feasibility describe the degree of difficulty associated with implementing a project. This includes engineering challenges such as long tunnels, deep geological formations, underwater structures, or unstable soil conditions; regulatory complexity such as multi-country permitting, environmental approvals, and alignment of planning frameworks; and administrative challenges such as coordination among infrastructure managers, national ministries, and regional authorities. Many past Trans European Transport Network projects have experienced long delays due to underestimated geological risks, cross-border legal discrepancies, or incomplete preparatory studies. Assessing feasibility helps identify projects that can realistically be completed within strategic timeframes, such as the 2030 core network deadlines or the 2050 extended network goals.
Reason for weight 4: Feasibility is essential because even highly beneficial projects may not succeed if their complexity is too great. However, it should not overshadow long-term strategic value, so it receives a balanced mid-range weight.
Definition for ELECTRE: The likelihood that the project can be delivered on time and within scope, considering technical, regulatory, administrative, and cross-border implementation risks.

2.5 Socio-economic benefits (weight: 9)

Socio-economic benefit is the most influential criterion because the primary purpose of the Trans European Transport Network is to strengthen the European Single Market, facilitate mobility, and promote economic, social, and territorial cohesion. These benefits include improved accessibility for citizens and businesses, reduced travel times, greater labour mobility, enhanced logistics performance, increased freight reliability, and better connections for remote or peripheral regions. Socio-economic benefits also encompass secondary effects such as job creation, tourism development, increased attractiveness for investment, and improved access to essential services. Because many of these benefits extend across borders, a single project can generate substantial positive spillovers for multiple member states. This aligns directly with long-standing European Union goals such as reducing regional disparities and supporting sustainable economic development.
Reason for weight 9: Socio-economic benefit is the central justification for European Union co-financing and therefore carries the highest weight. It reflects the expectation that Trans European Transport Network projects must generate broad and durable improvements for society and the economy.
Definition for ELECTRE: The degree to which the project delivers mobility gains, regional development, employment effects, accessibility enhancements, and positive economic spillovers at both national and European levels.

2.6 Safety and resilience (weight: 2)

Safety and resilience evaluate how well a project improves transport safety and the capacity of infrastructure to withstand disruptions. Safety aspects include modern signalling

systems, grade separation, reduced conflict points, and improved traffic management. Resilience includes measures that address climate change impacts such as flooding, heatwaves, landslides, storms, and other natural hazards, as well as risks related to geopolitical instability or supply chain disruption. Although these aspects are increasingly relevant under European climate and security policies, many modern transport projects must already meet high minimum standards for safety and environmental resilience. As a result, differences between projects in this category may be less pronounced unless a project directly mitigates a known high-risk area.
Reason for weight 2: Safety and resilience are essential but often do not differentiate strongly between competing projects, since all major European Union transport investments must comply with modern safety standards. The low weight reflects this baseline nature.
Definition for ELECTRE: The extent to which the project improves operational safety and enhances resilience against climate-related hazards, natural risks, and system disruptions.

3 Screening Analysis

The initial decision statement contains 10 alternatives that each need to be considered. However, it is important to filter out unimportant alternatives before continuing with deeper, more intensive analyses, so as not to waste resources. Since computational resources and time are scarce and valuable, it becomes counterproductive to perform every calculation for each possible alternative. Therefore, only alternatives with actual potential need to be considered.

This introduces the screening analysis. A screening analysis is a preliminary filtering stage, with the specific purpose of reducing a large set of alternatives to a smaller set of the most promising candidates. However, in the real world with many aspects to consider, this is easier said than done. To account for as many of these relevant factors as possible, the use of a multi-criteria decision analysis (MCDA) is required. An MCDA aims to model the multiple objectives and stakeholders that are involved in these projects.

Throughout history, multiple methods have been developed to apply an MCDA, one of these being the ranking method. This branch of MCDA ranks all the alternatives based on how they score on certain (weighted/hierarchical) criteria, enabling users to easily select the top alternatives to continue the analysis with. ELECTRE I is the oldest and simplest of these outranking methods.

3.1 ELECTRE technique

The ELECTRE technique is an MCDA ranking approach based on pairwise comparison, rather than a global overall score for each alternative. It does not use a hierarchy within the criteria but instead assigns weights to determine their importance. The ranking itself is done by comparing the concordance and discordance indices of each pair. The concordance index measures the support that the given information provides for the hypothesis that one option is at least as good as another. The discordance index shows the strength of evidence against this hypothesis. Before these characteristics can be calculated, the scores for each

alternative on the previously defined criteria must be determined. Table 1 shows these values on a 5-point scale.

Scenarios	Environmental Impact	Cross-Border Connectivity	Cost CAPEX + OPEX	Complexity/ Feasibility	Socio-Economic Benefits	Safety and Resilience
SC01	4	4	2	1	5	4
SC03	4	3	4	3	3	3
SC09	3	5	1	2	4	3
SC10	3	4	3	3	4	3
SC11	3	1	4	4	3	3
SC12	3	2	5	3	3	4
SC13	4	1	5	5	2	3
SC16	3	2	4	2	3	4
SC18	4	5	3	2	5	4
SC20	3	4	5	3	5	3
Weights	4	8	6	4	9	2

Table 1: ELECTRE Assigned Weights and Assigned 5-point scale where 1: VL, 2: L, 3: Av, 4:H, 5: VH

- **Concordance matrix:** Pairwise concordance indexes

  Applying the following formula: $C(a,b) = \frac{\sum_{i \in Q(a,b)} w_j}{\sum_{i=1}^{m} w_j}$, the concordance index for each pair is calculated. With $Q(a,b)$ being the set of criteria for which a is equal or preferred to b, and W_j the weight of criteria j. The results can be viewed in table 26 in the Appendix.

- **Discordance matrix:** Pairwise discordance indexes.

  Similarly to the previously mentioned concordance matrix, the discordance matrix visualizes all the pairwise discordance indices. However, a threshold needs to be defined, since the discordance indicates that if the score for an alternative on any criterion exceeds the score for another alternative on that criterion by an amount equal to or greater than its veto threshold, the second alternative can never outrank the first one. In this analysis, a threshold value of 2.5 on the 5-point scale was chosen. Using this value, the following formula is applied: $D(a,b) = \{1 \text{ if } z_i(b) - z_i(a) > t_i \text{ for any } i,\ 0 \text{ otherwise}\}$. The results can be viewed in table 27 in the Appendix.

3.2 Outranking Relation

Using the calculated concordance and discordance values from the previous section, the ranking itself can be performed. To visualize this ranking system, an outranking graph can be created (see Figure 1). Note that a threshold value for concordance is also defined here as 2,5. If an alternative A scores equal to or more than the threshold value compared to alternative B, A is preferred over B. If the difference between concordance scores is less than this threshold, the pairwise ranking will be undefined. In the figure, green arrows show an outranking and blue arrows show an undefined relation between two alternatives. Finally,

the results of this screening analysis can be seen in Table 4. Scenarios 1, 10, 18, and 20 come out on top of the ranking. Further analysis will therefore focus on comparing these four options.

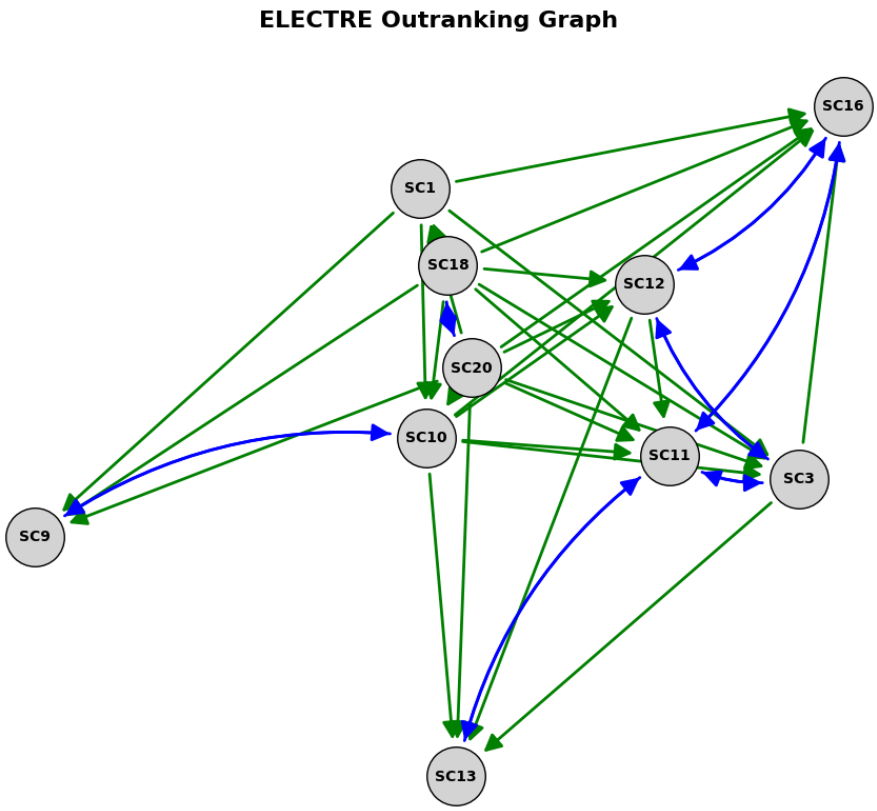

Figure 1: ELECTRE Outranking Graph

Top 4 Scenarios	Outranks	Outranked by	Net Flow
SC20	8	0	8
SC18	7	0	7
SC10	5	3	2
SC01	4	2	2

Table 2: Top 4 Scenarios based on ELECTRE Analysis

4 Cost-Benefit Analysis

The Cost–Benefit Analysis is carried out on the four scenarios identified in the screening stage and follows the standard socioeconomic appraisal framework prescribed for large transport infrastructure projects. All calculations are based on the dataset provided after

screening, including the scenario overview sheets, consumer surplus data, unit prices in the 2025 price level, and the baseline scenario (S00). Each scenario is assessed exclusively in terms of the difference it generates relative to this baseline, ensuring that all quantified impacts represent net societal changes rather than absolute levels.

The CBA includes all mandatory impact categories: construction and maintenance costs, scrap value, consumer surplus, and the principal externalities—air pollution, global warming, accidents, and noise. To structure the appraisal, first-year impacts for each of these components are computed using the modal breakdown of changes in traffic activity. For air pollution and global warming, monetary impacts are calculated separately for car petrol, car diesel, HGV diesel, passenger rail, freight rail, inland waterways, and air transport, after which they are aggregated into total category-specific values. Accidents and noise follow the same procedure but only for the modes where these impacts apply. These first-year impacts represent conditions in the opening year; therefore, forward or backward projection is not required at this stage.

Consumer surplus, which captures the combined benefits of reduced travel time and operating costs, plays a central role in the analysis. It is provided for the year 2035 in 2025 prices, already incorporating the rule-of-half to reflect the behavioural response of new users. Since the evaluation begins in the opening year of each scenario, these values are back-cast using the prescribed traffic and price growth rates. This ensures that the consumer surplus applied at year 0 reflects realistic demand and valuation conditions for the time of opening. Over the full evaluation period, consumer surplus is projected forward according to both traffic and price growth, consistent with its dual dependence on travel volume and value-of-time.

The remaining impact categories follow different forecasting profiles. Air pollution and global warming are adjusted using a negative emission growth rate, capturing the expected technological improvements and policy-driven reductions in emissions intensity over time. Maintenance costs, accident costs, and noise impacts remain constant, reflecting assumptions that increasing vehicle efficiency and safety will offset the effects of higher traffic volumes. All monetary values are discounted to the opening year using a 3% discount rate, which aligns the temporal profile of costs and benefits with the concept of time preference and the opportunity cost of capital. The evaluation horizon spans 50 years after construction is completed, and the residual value of the asset at the end of this period is included as a positive scrap value.

After projecting and discounting all components, the investment criteria are calculated for each scenario. The results, summarised in Table 3, show significant differences in economic performance.

Scenario	NPV Euro (Billion)	BCR	IRR %
SC01	11.000	0.33	0.60
SC10	6.121	1.54	4.54
SC18	0.299	1.04	3.13
SC20	1.089	1.40	4.23

Table 3: Cost Benefit Analysis Results for the four scenarios SC01, SC10, SC18 and SC20

SC01 generates a large absolute NPV driven by substantial benefit flows, yet because its investment cost is extremely high, its BCR falls to 0.33, indicating that benefits recover only a third of the discounted costs. Its IRR of 0.60% confirms that even under favourable discounting, the project fails to meet the minimum threshold for economic feasibility.

SC10 stands out as the strongest scenario, combining a high BCR of 1.54 with an IRR of 4.54%, both indicating a comfortable surplus of benefits over costs. Its balance between sizeable consumer surplus gains and relatively moderate investment requirements results in a consistently positive net return across the evaluation period.

SC18, with a BCR of 1.04, represents a marginally viable scenario. Its narrow buffer above the threshold of 1 suggests that the project is only just able to offset its costs, leaving little room for variations in demand, costs, or externalities. This sensitivity is also reflected in its IRR of 3.13%, which only slightly exceeds the discount rate.

SC20 performs robustly with a BCR of 1.40 and an IRR of 4.23%, supported by stable and growing consumer benefits relative to its cost profile. Although its NPV is lower than SC10's, the scenario demonstrates solid economic value and a strong ability to generate net societal gains.

Taken together, the CBA identifies SC10 and SC20 as the most economically promising options, offering clear net benefits and resilient benefit–cost structures. SC18 remains feasible but with limited margin for uncertainty, while SC01 despite its large benefit volume proves economically unsound due to disproportionate investment requirements. These findings provide the foundation for the subsequent risk analysis and strategic assessment.

5 Feasibility Risk Assessment

The FRA is performed on the CBA of the 4 scenarios using Monte Carlo Simulation (MCS). The purpose of the FRA is to examine the impact the demand forecast (consumers surplus) and construction cost have on the final benefit-cost ratio (BCR). The following are the probability distributions applied to the scenarios with regards to construction cost and consumer surplus:

- **Consumer Surplus:** Beta-PERT Distribution

  Given a minimum, mean, and maximum of -92.3%, -37.0%, and 144.2% respectively for rail projects, as well as -78.5%, 9.6%, and 179.34% respectively for road projects, table 4 shows the input values for the Beta-PERT distribution for either mainly road or rail projects.

	a	b	c	α	β
Rail	0.077	0.630	2.442	1.935	4.065
Road	0.215	1.096	2.793	2.367	3.633

Table 4: Beta-PERT distribution inputs based on either mainly rail or road projects

- **Construction Costs:** Gamma (Erlang) Distribution

	k	θ
Rail	23.00	0.075
Road	8.00	0.09

Table 5: Gamma distribution inputs based on either mainly rail or road projects

Scenario 1 is clearly defined as a railway project in the full project data provided, therefore the rail probability distribution inputs are applied to scenario 1. The other three scenarios are a combination of rail and road works, as such it is important to identify whether these scenarios are mainly road or rail works.

- **SC01: Mainly railway**, as stated in the full project data.
- **SC10: Mainly railway** since the total length of railwork (1833 km) is significantly greater than total roadwork length (456 km)
- **SC18: Mainly railway**, since the bulk of the work is dedicated to upgrading and construction of the railway line, with a new 848 km railway upgrade. The road component is secondary to the railway network expansion
- **SC20: Mainly roadwork**, since the total length of the roadwork (625 km) is almost twice the length of railwork (376 km). The quantified work heavily favors upgrading existing roads.

For each scenarios a 2000 iteration Monte-Carlo simulation is conducted by calculating the BCR at each iteration and only varying the construction cost (and subsequent scrap value) and the consumer surplus with the previously defined probability distributions. Figure 2 is the BCR distribution result of the 2000 iterations for each of the 4 scenarios. The mean, standard deviation, minimum and maximum values of each scenarios can be found in table 22 in the Appendix.

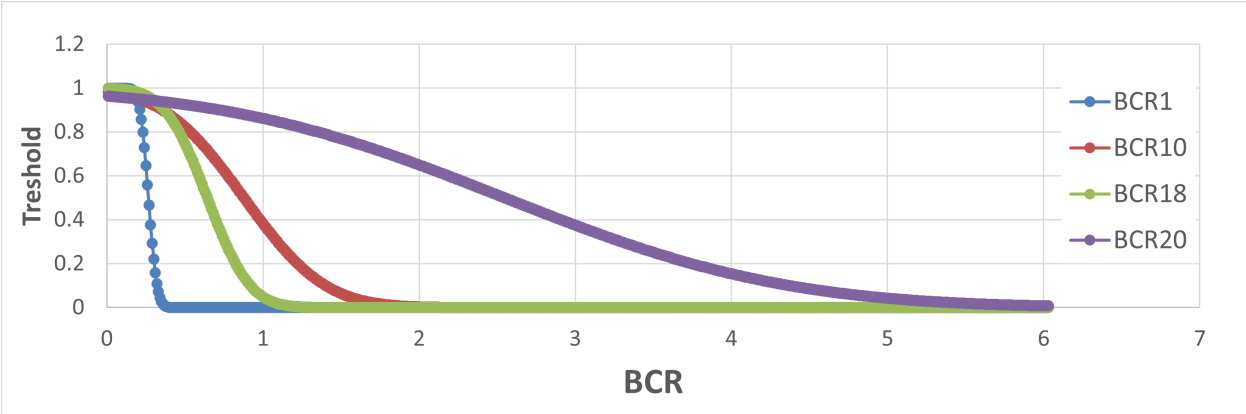

Figure 2: BCR Cumulative Distribution of 2000 samples for all 4 scenarios

The result of the FRA reveal a distinct difference in the projected profitability of the scenarios based on their BCR distributions. SC20 and SC10 demonstrate favorable distributions, as a BCR exceeding 1 is the critical threshold for financial viability, indicating that the benefits outweigh the costs. Specifically, SC20 shows exceptional financial appeal, with the majority of its distributed BCR values ranging between 1 and 6, suggesting highly profitable results. Although not as consistently high as SC20, SC10 maintains some likelihood of profitability, as a portion of its BCR distribution is also greater than 1. On the other hand, SC1 and SC18 face greater feasibility risks. Their distributions are primarily composed of BCR values less than 1, meaning these scenarios have a higher probability of incurring costs that exceed the anticipated benefits, indicating lower overall profitability and financial prudence.

6 Multi-Criteria Decision Analysis

The Multi-Criteria Decision Analysis (MCDA) is a structured method used to evaluate and compare alternatives based on multiple criteria. The analysis encompasses quantitative and qualitative information, and it is especially useful when decisions involve trade-offs between factors that are difficult to directly compare. The MCDA focuses on impacts that cannot be easily monetized and is outside the scope of the CBA, the criterias chosen for this section cannot overlap with any of the impacts and factors described in the CBA. The MCDA is also a refinement built upon the previous ELECTRE analysis, with the difference being that the criteria are more well defined and does not overlap one another. Below are the following chosen criteria focusing on the economic, environmental and social aspects of the project scenarios outside of the CBA:

- Economic:
  - **C1:** Integration of European transport
  - **C2:** Location of Companies and Logistic Centers
- Environmental:

  - **C3:** Ecosystem and Biodiversity Loss
  - **C4:** Non-Renewable Dependence

- Social:
  - **C5:** Labour Market and Sociographic Job Distribution
  - **C6:** Territorial Cohesion

The method chosen for the MCDA is a pairwise method called REMBRANDT (Ratio Estimation in Magnitudes or Decimals Technique). The pairwise method is chosen over the SMART method in this case because of consistency, group decision making and less cognitive load. Pairwise approaches allows for consistency checks, where weights are assigned based on direct comparisons between alternatives, meaning that the ranking between comparisons are always consistent. SMART approach on the other hand cannot test for internal logic, the weights assigned from value functions may be arbitrary or at worst contradictory. The SMART technique also requires estimating exact weights based on proven or established value functions that may or may not exist. In the case for the TEN-T projects, accurate and comparable information on the mentioned criteria are difficult to source. For example, accurate estimation of non-renewable emission level would be required to assess criteria C4 in which the method to calculate such emissions would require more cognitive load and methods outside the scope of this project. Finally, the last main advantage of direct pairwise comparison is the opportunity for group decision making, allowing for discussions between how alternatives rank amongst the group, which reduces bias and facilitates averaging of judgements. SMART analysis on the other hand can hide inconsistencies between experts.

The REMBRANDT method was chosen over the Analytic Heirarchy Process (AHP) method due to its intuitiveness and robustness to inconsistency. REMBRANDT uses a logarithmic magnitude scale which alligns better with how decision makers perceive intensity. AHP consistency index requirement requires rework to ensure consistency while REMBRANDT does not require this step. AHP can potentially suffer from rank reversal, which is the change in alternative ranking when an option is added or removed. REMBRANDT has greater stability because of its ratio-based calculation.

Each alternatives (SC1, SC10, SC18 and SC20) are compared to one another based off the aforementioned criteria and a comparison matrix is created. S_k refers to the scenarios at the horizontal portion of the matrix, while S_j refers to the scenarios at the vertical portion of the matrix. A gradation index is assigned based on table 6 for each comparisons. Each value in the matrix is converted into the geometric scores $S_{j,k}^{geo}$ using equation 1 . At each row the geometric mean for each scenarios is calculated using the *GEOMEAN* function in excel.

$$S_{j,k}^{geo} = \exp(LN(\sqrt{2})) \times \delta_{j,k} \tag{1}$$

Comparative Judgement	Gradation Index $\delta_{j,k}$
Very strong preferencec for S_k over S_j	-8
Strong preferencec for S_k over S_j	-6
Definite preferencec for S_k over S_j	-4
Weak preferencec for S_k over S_j	-2
Indifference	0
Very strong preferencec for S_j over S_k	+2
Strong preferencec for S_j over S_k	+4
Definite preferencec for S_j over S_k	+6
Weak preferencec for S_j over S_k	+8

Table 6: REMBRANDT Comparative Judgement Scores

6.1 C1: Integration of European transport

The first criterion in this section compares the alternatives based on the integration of European transport. The alternatives fall into two categories: creating new routes or upgrading existing ones. If a project aims to upgrade an existing route, the current use of this route is used in the comparison, specifically the amount of cargo and private transport that travels through it. When a new route is created, the calculations need to be adjusted. Since there is no existing route that can be directly used to quantify the amount of cargo and private transport, the closest match is used. In other words, the most common route between the starting point and endpoint of the envisioned new connection is used to quantify the relevant numbers.

The estimates are as follows: Sc1 with 11.5 million passenger cars and 40 million tonnes of freight, Sc10 with 1.44 million cars and 15.26 million tonnes of freight, Sc18 with 1.2 million cars and 56 million tonnes of freight, and finally Sc20 with 15 million cars and 56.16 million tonnes of freight. Since this is an economic benchmark, the amount of freight carries a greater weight. The preference results can be found in the table below.

	SC1	SC10	SC18	SC20	Geo Mean
SC1	0	4	-2	-4	0.707
SC10	-4	0	-6	-6	0.063
SC18	2	6	0	-2	2.828
SC20	4	6	2	0	8.000

Table 7: REMBRANDT Comparison Matrix for Criteria 1

6.2 C2: Location of Companies and Logistic Centers

Focusing on the economic side, this criterion also shows preferences for the alternatives based on economic activities. However, rather than focusing on the activity along the route, it looks at the economic activity of the cities and/or regions that will be connected. Quantification was done based on the gross domestic product (GDP) of the relevant places. The

total GDP of the connected regions is as follows: Sc1 connects a total of 169.5 billion in GDP, Sc10 affects 23.87 billion, Sc18 corresponds to 305.43 billion, and finally Sc20 has a total of 241 billion.

	SC1	SC10	SC18	SC20	Geo Mean
SC1	0	6	-6	-4	0.500
SC10	-6	0	-8	-8	0.022
SC18	6	8	0	2	16.00
SC20	4	8	-2	0	5.657

Table 8: REMBRANDT Comparison Matrix for Criteria 2

6.3 C3: Ecosystem and Biodiversity Loss

For the environmental side, Table 9 summarises the REMBRANDT comparisons for the criterion *Ecosystem and biodiversity loss*. The ratings are based on the total length and type of works in each Sc, since new linear infrastructure is assumed to cause more habitat loss and fragmentation than upgrades within existing corridors. Sc10 performs worst, with 456 km of new roads and 52 km of new rail, implying extensive new land take. Sc1 has 55 km of new tunnel railway and long rail upgrades, which still generate substantial disturbance but less new surface fragmentation than Sc10. Sc18, with only 19 km of new roads and 19 km of new rail plus rail upgrades, is judged moderately better. Sc20, which consists almost entirely of road and rail upgrades, is considered best as it mainly reuses existing rights-of-way. Table 9 reflects this ordering, with larger positive values assigned when an Sc has clearly lower ecological impact than the comparison alternative.

	SC1	SC10	SC18	SC20	Geo Mean
SC1	0	2	-4	-6	0.250
SC10	-2	0	-6	-8	0.063
SC18	4	6	0	-4	2.828
SC20	6	8	4	0	22.63

Table 9: REMBRANDT Comparison Matrix for Criteria 3

6.4 C4: Non-Renewable Dependence

Table 10 reports the REMBRANDT matrix for the criterion *Non-renewable dependence*. Here the logic focuses on the balance between road and rail works, since rail, specially electrified rail, is assumed to reduce long-term dependence on oil relative to road transport. Sc1 is rated best because it consists almost exclusively of new and upgraded rail, favouring a shift towards a mode that can be powered by electricity from renewable sources. Sc18 mixes limited new roads with new rail and upgrades, offering a smaller but still positive shift towards rail. Sc20 mainly upgrades existing roads and rail; this may slightly improve efficiency but does not significantly change the modal split, so it is placed below Sc18. Sc10 is rated worst because its large programme of new roads is likely to lock in future oil-dependent

car and truck traffic. Table 10 encodes these differences, with stronger preferences where the shift away from road and towards rail is more pronounced.

	SC1	SC10	SC18	SC20	Geo Mean
SC1	0	6	4	4	11.31
SC10	-6	0	-4	-2	0.125
SC18	-4	4	0	2	1.414
SC20	-4	2	-2	0	0.500

Table 10: REMBRANDT Comparison Matrix for Criteria 4

6.5 C5: Labour Market and Sociographic Job Distribution

C5 compares alternatives based on the potential number of jobs created by the project. The total kilometers of new rail works, rail maintenance, new road works, road maintenance, and specific unique infrastructure (such as tunnel work or bridge work) is calculated for each scenario. Projects with unique infrastructure have at least strong preference over project without since those projects would usually require more diverse expertise and manpower to construct. According to an article by *Transportation for America* on road and public transit maintenance [2], road maintenance produces 16% more jobs than new roads, while public transport works created 70% more jobs than road works. An article by Smart Growth America also reveals that public transportation work could create 31% to 70% more jobs than road works, based on the analysis done by 2009's American Recovery and Reinvestment Act [3]. Unfortunately no other concrete data was found when comparing road and rail works in terms of job creations. But based on this logic, the total kilometers of the road and rail works can be summed up for each scenarios and a weight between 1.3 and 1.7 can be assigned to railworks to directly compare the job creations in terms of roadworks. Applying this logic allows the decision makers to objectively compare the alternatives based on the number of jobs alternatives with no unique infrastructure can create. Table 11 shows the resulting comparison matrix from applying the aforementioned logic.

	SC1	SC10	SC18	SC20	Geo Mean
SC1	0	4	-4	6	2.828
SC10	-4	0	-2	6	1.000
SC18	4	2	0	6	8.000
SC20	-6	-6	-6	0	0.044

Table 11: REMBRANDT Comparison Matrix for Criteria 5

6.6 C6: Territorial Cohesion

Territorial Cohesion has many different definitions, but for simplicity and ease of objective comparison, a territorial cohesion index (TCI) can be applied to each countries where each scenario passes and an average TCI can derived. Importance is derived based on a scenario's average TCI. Scenarios with lower average TCI should be prioritized since the TEN-T

project is used to address areas with lower territorial cohesion. Research by Medeiros, E., et. al on *Measuring territorial cohesion trends in Europe: a correlation with EU Cohesion Policy* shows the TCI based on EU territory [4]. Using the TCI values provided in figure 1 of the report results in the comparison matrix table 12 for C6.

	SC1	SC10	SC18	SC20	Geo Mean
SC1	0	2	4	6	8.000
SC10	-2	0	4	6	4.000
SC18	-4	-4	0	6	0.707
SC20	-6	-6	-6	0	0.044

Table 12: REMBRANDT Comparison Matrix for Criteria 6

6.7 Criteria Weights

Pairwise comparisons of the criteria are done to derive the weights of each criteria. The weights of each criteria are derived using the equation 2, where the geometric mean of the scenario is divided by the total sum of geometric mean for all scenarios.

$$w_n = \frac{GEOMEAN(SC_n)}{\sum_{n=1}^{N} GEOMEAN(SCn)} \tag{2}$$

The resulting comparison table 13 derived from group discussion, heavily prioritizes the economic criteria over the environmental and social criteria. The TENT-T project prioritizes sustainability, connectivity and efficiency meaning that all criteria listed help achieves the goal set by the European Union. However, after discussion, the resulting argument is that the project should prioritize infrastructure investments that maximizes seamless pan-European trade and logistic performance (C1 and C2). Environmental and social impacts should be managed and mitigated, but they should not be the decisive factors in route selection or project approval unless the economic benefits are nearly equal.

	C1	C2	C3	C4	C5	C6	Geo Mean	Weights
C1	0	2	4	6	6	6	4.000	0.459
C2	-2	0	2	4	4	6	2.245	0.258
C3	-4	-2	0	2	2	2	1.000	0.115
C4	-6	-4	-2	0	2	2	0.630	0.072
C5	-6	-4	-2	-2	0	0	0.445	0.051
C6	-6	-6	-2	-2	0	0	0.397	0.045

Table 13: REMBRANDT Comparison Matrix for the Six Criterias and Their Resulting Weights

6.8 Sensitivity Analysis: Stakeholder Viewpoints

To assess the result's robustness towards changes in criteria weights, three additional viewpoints were explored:

- **Environmentalist:** The environmentalist viewpoint matrix (Table 23 in the Appendix) argues for an Ecology-first strategy. Economic integration and logistic efficiency would only be pursued within strict environmental constraint imposed by the two dominant criteria C3 and C4.
- Labor Workforce: The workforce viewpoint matrix (Table 24 in the Appendix) shows a heavy focus on social factors, particularly labor market conditions, followed by a prioritization of the economic elements that enables job creation and accessibility.
- Corporations: The corporation viewpoint matrix (Table 25 in the Appendix) reveals a clear strategy focused on maximizing business logistics efficiency and market access. The highest priority is given to elements that reduce friction and cost for companies operating across the network.

Figure 3 shows the resulting MCDA criteria weights under different stakeholder viewpoints resulting from the stakeholder criteria matrix. The figures shows that the resulting weights from the stakeholder analysis is aligned with the previously defined viewpoints. Environmentalist heavily favors the environmental criteria, corporations heavily favors the company locations and logistics while the workforce heavily favors the labor market and sociographic distribution.

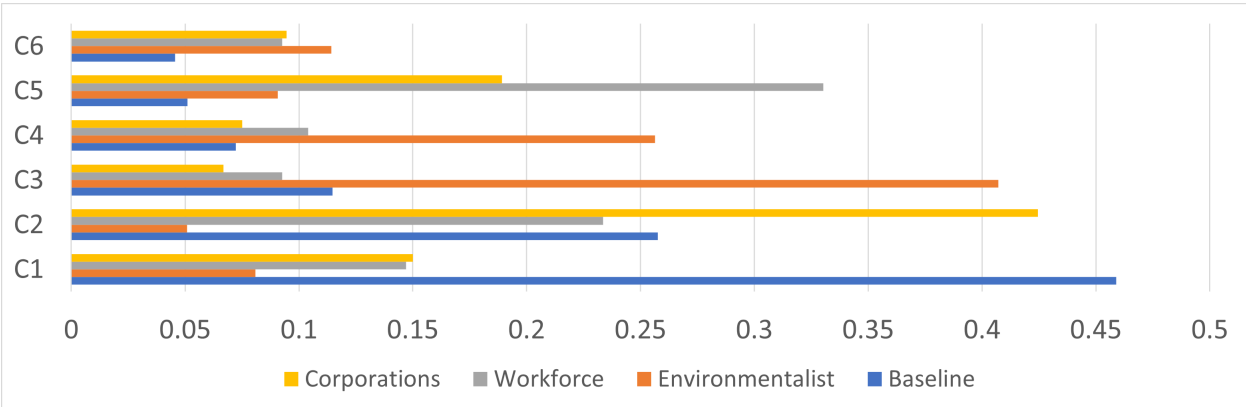

Figure 3: MCDA Criteria Weights under Different Stakeholder Viewpoints

6.9 Results

Figure 4 shows the resulting normalized REMBRANDT scores based on the criteria weights in figure 3. Ranking the scenarios based on their REMBRANDT scores reveals a some common and distinct trends among scenarios. SC10 is consistently ranked the bottom across all scenarios, meaning that SC10 is the least susceptible towards criteria weights

Priority	Baseline	Environmentalists	Workforce	Corporations
1	SC20	SC20	SC18	SC18
2	SC18	SC1	SC20	SC20
3	SC1	SC18	SC1	SC1
4	SC10	SC10	SC10	SC10

Table 14: Priority Table Based on Normalized REMBRANDT Scores

(Table 14). Workforce and Corporations have the exact same ranking, highlighting that the needs of the workforce aligns with the needs of the corporations. Both groups favor a scenario that maximizes economic and logistical efficiency. Subtle differences in their preferred strategic approach did not alter any of their preference. SC20 and SC1 represent scenarios that inveolves significant shifts towards sustainability. While SC1 is ranked low by other stakeholders, it is only truly embraced by the environmentalist viewpoint. SC18 and SC20 dominates the top two positions across the four groups, with SC20 also appearing as the top choice for the environmentalist. This strongly suggest that a hybrid or compromise between SC18 and SC20 would provide the broadest appeal and consensus among the most influential stakeholders.

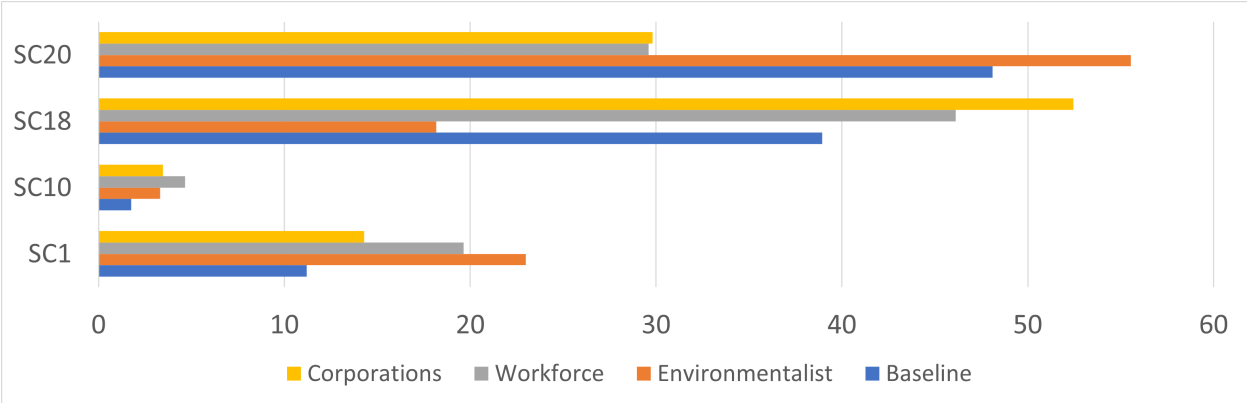

Figure 4: MCDA Normalized Scores under Different Stakeholder Viewpoints

7 Composite Assessments

Composite assessments combine both results from the CBA and the MCDA. Two composite assessment menthods are explored in this report, the SIMDEC method and the COSIMA method.

7.1 SIMDEC Method

The SIMDEC method adds an additional criteria to the MCDA titled "Socioeconimic Robustness". The Socioeconomic Robustness is a way to combine the Monte Carlo results

from the FRA with the MCDA method. The FRA results shown in figure 2, shows a clear BCR distribution across scenarios. Scenarios with a higher distribution of BCR > 1 is more preferable over scenarios with more distribution in lower BCR range. SC20 has the highest spread of BCR distribution with most landing above 1, meaning that SC20 is the most prefered among the four scenarios. SC1 has a BCR distribution concentrated below 1 meaning that it performs the worst among the four. Based on this logic the REMBRANDT comparison matrix for criteria 7 (C7) can be generated as shown in table 15.

	SC1	SC10	SC18	SC20	Geo Mean
SC1	0	4	4	-7	1.189
SC10	-4	0	4	-6	0.354
SC18	-4	-4	0	-6	0.088
SC20	7	6	6	0	26.91

Table 15: REMBRANDT Comparison Matrix for additional Criteria 7: Socioeconomic Robustness

The cost and benefit of the project is definitely preferred over the criteria addressed in the MCDA. This is because the criteria is only secondary to the cost and benefit of the project, without proper cost and benefit assessment, the fundamental feasibility of the project could not be properly assessed. Table 16 shows the resulting new REMBRANDT criteria matrix, where C7 is definitely preferred across all existing criteria.

	C1	C2	C3	C4	C5	C6	C7	Geo Mean	Weights
C1	0	2	4	6	6	6	-4	2.692	0.276
C2	-2	0	2	4	4	6	-4	1.641	0.168
C3	-4	-2	0	2	2	2	-4	0.820	0.084
C4	-6	-4	-2	0	2	2	-4	0.552	0.057
C5	-6	-4	-2	-2	0	0	-4	0.410	0.042
C6	-6	-6	-2	-2	0	0	-4	0.371	0.038
C7	4	4	4	4	4	4	0	3.281	0.336

Table 16: REMBRANDT Comparison Matrix for the Six Criterias and Their Resulting Weights

The resulting new normalized REMBRANDT score (Figure 5) shows a larger score and emphasis on SC20. Thus, when accounting for the cost and benefit with the FRA data, the favorable BCR result of SC20 increases the weight and importance of SC20 to the decision making process. The scenario rankings do not change, rather it emphasizes SC20 dominance over the other scenario due to the support of the BCR result.

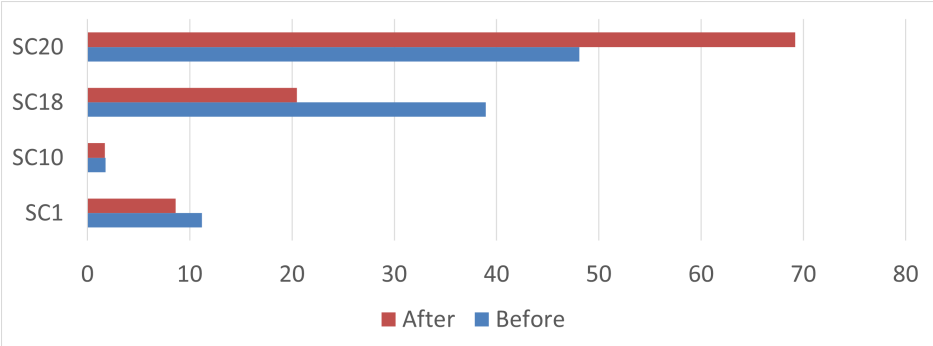

Figure 5: SIMDEC Result - Total Normalized REMBRANDT Scores Before and After Additional Socioeconomic Robustness Criteria

7.2 COSIMA Method

Unlike the SIMDEC approach, the COSIMA method combines the CBA result with the MCDA results, and not the FRA results. First the Geometric mean of each criteria (C1 to C6) is converted to a value ranging from 0 to 100 using the value function 3 with a linear assumption, where $V_{i,j}$ is the value function result for criterion i and scenario j. The variables $G_{i,j}$, $G_i^{\min}$, and $G_i^{\max}$ represent, respectively, the geometric mean value for criterion i and scenario j, the minimum geometric mean value observed across all scenarios for criterion i, and the maximum geometric mean value observed across all scenarios for criterion i. Table 17 shows the value function results applied to the geometric mean matrix.

$$\text{Value}_{i,j} = \frac{G_{i,j} - G_i^{min}}{G_i^{max} - G_i^{min}} \times 100 \tag{3}$$

	SC1	SC10	SC18	SC20	Weight
C1	8.121	0.000	34.85	100.0	0.459
C2	2.991	0.000	100.0	35.27	0.258
C3	0.831	0.000	12.26	100.0	0.115
C4	100.0	0.000	11.52	3.352	0.072
C5	35.00	12.01	100.0	0.000	0.051
C6	100.0	49.72	8.332	0.000	0.045

Table 17: Value Function Results from REMBRANDT Geometric Mean using Linear Assumption

For each MCDA% the following calculations are performed. First a scenario is chosen as a calibration point, usually the scenario with the highest BCR is chosen, which in this case is SC10. First the MCDA Benefit $MCDA_b$ is derived using equation 4, where B_b is the benefit amount from the chosen calibration scenario.

$$MCDA_b = B_b \frac{MCDA\%}{1 - MCDA\%} \tag{4}$$

The following set of linear equations (Equations 5 and 6) is then solved to derive the shadow prices for each respective criterion (sp_j), where $sp_{j=1}$, and $w_{j=1}$ are the respective shadow price and weight for the first criteria. V_{bj} are the values for the calibration scenario b and the criterion j. Table 18 shows the shadow prices calculated for the 70% MCDA% case using this method.

$$\sum_{b,j} V_{bj} sp_j = MCDA_b \tag{5}$$

$$sp_j = \frac{w_j}{w_{j=1}} sp_{j=1} \tag{6}$$

Shadow Prices	sp_1	sp_2	sp_3	sp_4	sp_5	sp_6
MDKK	6482	3638	1620	1021	722	643

Table 18: Shadow Prices sp_n for MCDA% of 70%

Once the shadow prices are calculated, the TRR for each scenario i can be calculated using equation 7, where B_i and C_i are the respective benefit and cost of scenario i. Table 19 shows the TRR result for the 70% MCDA% case using this method.

$$TRR_i = \frac{B_i + \sum V_{i,j} sp_j}{C_i} \tag{7}$$

Scenario	SC1	SC10	SC18	SC20
TRR	15.85	5.139	95.67	344.8

Table 19: TRR for MCDA% of 70%

The shadow price and TRR calculation process is repeated for a range of MCDA% values between 0% and 70%. Figure 6 is the resulting graph of the calculations. The plot for SC18 and SC20 appears to be unrealistically large, reaching the hundreds scale, which should not be the case. This result suggests that the calibration point chosen leads to an unrealistically high weighing of the MCDA scores. The values for the calibration scenario are mostly 0 (for criterion C1 to C4), while the values for SC18 and SC20 are dominated by 100 and overall higher scores. This significant gap in values between the chosen calibration scenario and the rest of the scenarios results in unrealistically high TRR results shown in figure 6. No meaningful analysis can be conducted using this plot; however, if the calibration point is shifted to the next scenario with the highest BCR, this problem could be resolved.

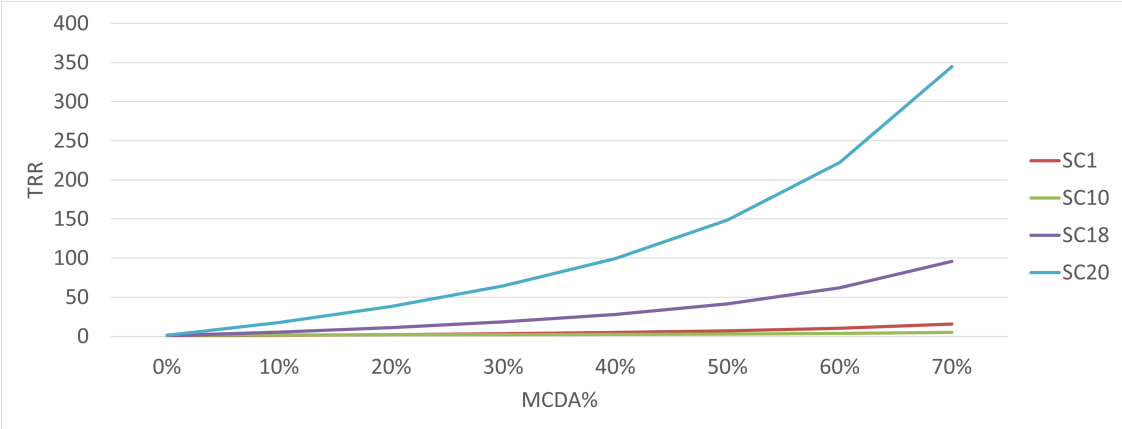

Figure 6: COSIMA Result - TRR Vs. MCDA% with Scenario 10 as Calibration Reference

Shifting the calibration point to SC20 as the scenario with the second largest BCR, results in the graph shown in figure 7. This decision should be acceptable, since the goal of the COSIMA calibration is to find a meaningful trade-off factor between the CBA and MCDA components. Since the previous result is unstable, illogical and not suitable for analytical purposes, it is best to shift the calibration point since using the highest BCR scenario is not a hard, immutable rule. The new result is much more stable with no extremely high TRR points.

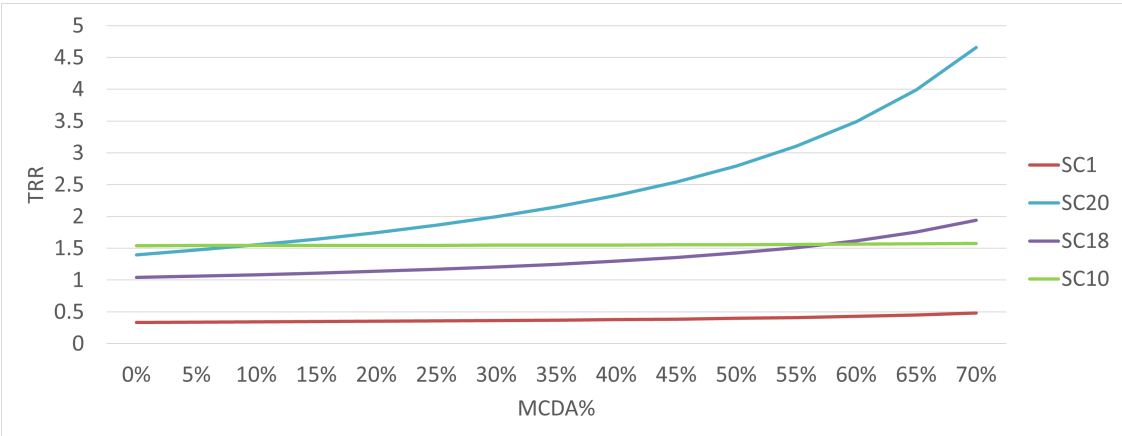

Figure 7: COSIMA Result - TRR Vs. MCDA% with Scenario 20 as Calibration Reference

MCDA% represents the importance given to the criterion defined in the MCDA section; a higher MCDA% means that decision makers are more willing to sacrifice pure economic efficiency (BCR) to achieve the strategic objectives outlined in the MCDA. The results show that when purely looking at the projects in terms of cost and benefit, SC10 remains the most attractive. However, when additional strategic objectives begin to be prioritized, the importance of SC10 begins to diminish. SC10 scored poorly when it comes to achieving the MCDA criterion, which means that no matter how large MCDA% gets, the TRR remains stable since the influence that MCDA has on SC10 is relatively small compared to the rest of

the scenarios. SC20 and SC18 show the greatest improvements when accounting for strategic objectives, SC20 in particular already exceeds the TRR score of SC10 as soon as it exceeded 10% in MCDA%. SC18, on the other hand, only exceeds SC10 in terms of TRR towards the end of the MCDA%, which means that the importance of the strategic objective must exceed the economic importance for SC18 to be more attractive than SC10. SC20 experiences the highest rate of increase in TRR, which means that it becomes significantly more attractive the more strategic objectives are considered. Thus, in terms of the COSIMA analysis, SC20 is the most attractive scenario since it does not require a large economic sacrifice to achieve the strategic goal.

8 Sensitivity Analysis

For the final analysis, a sensitivity analysis is conducted for the REMBRANDT results by varying the progression factor (γ) from 1 to 2.7. The analysis examines the influence of fundamental scaling choice on the final ranking alternatives. The progression factor symbolizes the intensity of preference, increasing the factor increases the magnitude of preference attributed to the verbal judgments. A higher factor means a judgment translates into a much higher numerical ratio of preference causing the the scores of the best alternatives to be further separated from the scores of the average or poor alternatives. Figure 8 shows the REMBRANDT score of the varied progression factor on the alternative score level (γ values between 1 and 2.7). The result shows that all the scenarios maintained their position and thus the ranking is considered robust and less sensitive to a specific scaling choice.

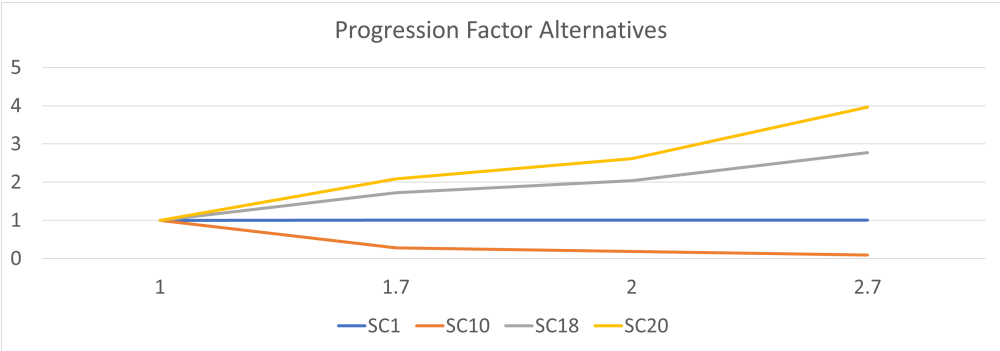

Figure 8: Sensitivity Analysis - Variation of Progression Factor (x-axis) Vs REMBRANDT Score (y-axis) for Alternatives

The sensitivity analysis done by varying γ of the criteria weight on the other hand shows some rank reversals, where SC20 continues to go up in rank as γ increases. The more decision makers let their favored alternatives pull away from the average ones, the higher the SC20 REMBRANDT score gets. The REMBRANDT scores of the other scenarios on the other hand remains stable with similar seperation regradless of γ.

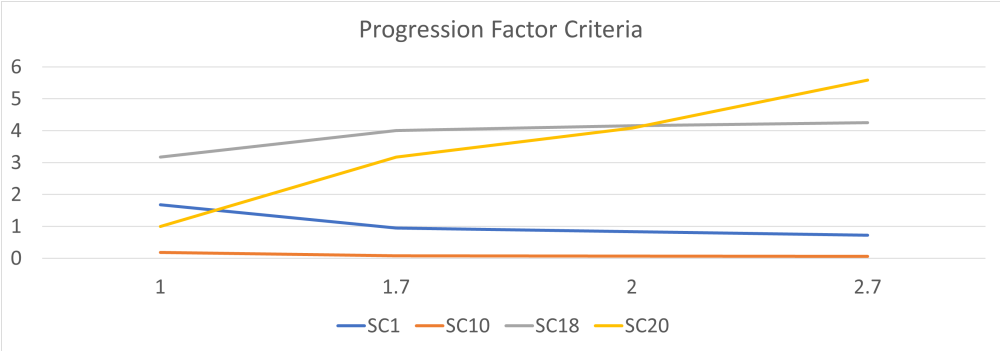

Figure 9: Sensitivity Analysis - Variation of Progression Factor (x-axis) Vs REMBRANDT Score (y-axis) for Criteria Weight Level

9 Discussion and Conclusion

Combining the results from the CBA, FRA, and MCDA using the composite assessments gives the decision maker a more comprehensive view for decision making.

Starting from the initial 10 scenarios, an ELECTRE analysis was performed to filter out the non-important alternatives. This resulted in a selection of the four most promising candidates: Sc1, Sc10, Sc18, and Sc20. These claimed the top spots in the ELECTRE ranking. Thus, further analysis was conducted on these four scenarios.

First of all, a cost-benefit analysis (CBA) was performed. This method aims to quantify the benefits a project would generate. By putting everything in monetary terms, the projects can easily be compared to each other. This analysis resulted in a net present value (NPV) and an internal rate of return (IRR) overview. Here, Sc1 emerged with the highest NPV in absolute value; however, the IRR showed a different story. Sc1 scored by far the lowest, with the other three projects being closely matched.

To complement the CBA, a feasibility risk assessment was also conducted. This assessment used a Monte Carlo simulation to examine the impact that demand forecasts and construction costs have on the final result. Here, Scenario 20 stood out with an extremely favorable distribution. Scenario 10 also showed favorable results, followed by Scenario 18, which had a harder time. Scenario 1 showed troubling signs, with profitability thresholds far below the others and less than 1.

The SIMDEC results show that incorporating socioeconomic robustness, derived from the FRA results, further strengthens the dominance of SC20, which already exhibits the widest spread of BCR values above 1 and therefore demonstrates the most robust welfare performance among the four scenarios. SC1, by contrast, has BCR values concentrated below 1, confirming it as the least economically attractive option, while SC10 and SC18 remain in between but clearly outperformed by SC20 once the MCDA criterion is added. Because cost and benefit are given more weight and importance when compared to other MCDA criteria, the SIMDEC weighting scheme systematically shifts additional decision weight toward SC20 without altering the original ranking order.

The COSIMA analysis results reinforces the SIMDEC findings by quantifying how

project attractiveness changes as decision makers place more emphasis on the strategic MCDA objectives. Using SC10 as the calibration point leads to unstable and unrealistically high TRR values, driven by extreme differences in value scores between SC10 and the other scenarios such as SC20 and SC18. Recalibrating on SC20 produces a stable TRR profile and shows that SC20 overtakes SC10 at low MCDA% values and continues to gain attractiveness as strategic objectives become more important, whereas SC18 only becomes preferable to SC10 at high MCDA% levels and SC1 remains unattractive throughout the MCDA% range.

Taken together, the SIMDEC and COSIMA results indicate that SC20 should be recommended as the preferred scenario, as it combines robust cost-benefit performance with strong alignment to strategic MCDA criteria across a wide range of decision-maker preferences. SC18 may be considered only in cases where strategic objectives are explicitly prioritized above economic efficiency and when SC20 becomes unavailable as an option. SC1 should be ruled out given its systematically poor performance across all metrics discussed besides the NPV value. However, NPV value is negated by the low IRR which makes SC1 unattractive as a whole. Decision makers should therefore focus further design and stakeholder dialogue on SC20. Further design work on SC20 should explore technical refinements that preserve the strong BCR and MCDA performance while testing its robustness under alternative demand, cost, and policy assumptions. Stakeholder dialogue can be structured through workshops to validate criteria, discuss trade-offs and reveal impacts that may not be fully captured in the quantitative MCDA model. Engaging key political, community and expert stakeholders in the SIMDEC and COSIMA outcomes for SC20 helps build ownership and trust towards the perefered option, reducing conflict and increasing likelihood of successful implementation.

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10 Appendix

10.1 Contributions

Name Section/student	Muhammad Alif s242524	João s250263	Dante s252291	Levente s242960
Calculations:				
Screening Analysis	25%	25%	25%	25%
Socioeconomic Analysis	25%	25%	25%	25%
Feasibility Risk Assessment	25%	25%	25%	25%
Strategic Analysis	25%	25%	25%	25%
Composite Assessment	25%	25%	25%	25%
Report:				
Screening and Definition	25%	25%	25%	25%
Screening Analysis	25%	25%	25%	25%
CBA	25%	25%	25%	25%
FRA	25%	25%	25%	25%
MCDA	25%	25%	25%	25%
Composite Assessment	25%	25%	25%	25%
Sensitivity Analysis	25%	25%	25%	25%
Results and Discussion	25%	25%	25%	25%

Table 20: Table of contributions

10.2 Declaration for the use of AI

Large language models and GenAI such as ChatGPT, Gemini, and Perplexity were used during the report writing process to create rough drafts, ideas and grammar checks. GenAI were also used to ask for information and help when TAs were not available for help to not hinder the progress of the project, later these informations were crosschecked by the TAs to confirm their validity.

10.3 Tables

Scenarios	TEN-T Priority Projects	Scenarios	TEN-T Priority Projects
1	PP1	12	PP12
3	PP3, PP8, PP19	13	PP12
9	PP6, PP22	16	PP29
10	PP7, PP22	18	PP20
11	PP7, PP22	20	PP24

Table 21: Scenarios to be examined and their corresponding TEN-T Priority Projects

Scenario	Mean	Std. Dev	Min	Max
SC01	0.266	0.043	0.192	0.474
SC10	0.878	0.408	0.201	2.845
SC18	0.644	0.215	0.283	1.708
SC20	2.517	1.422	0.504	13.039

Table 22: Benefit Cost Ratio (BCR) results - 2000 samples Monte-Carlo Analysis

	C1	C2	C3	C4	C5	C6	Geo Mean	Weights
C1	0	2	-4	-4	0	-2	0.630	0.081
C2	-2	0	-6	-6	0	-2	0.397	0.051
C3	4	6	0	2	4	4	3.174	0.407
C4	4	6	-2	0	2	2	2.000	0.256
C5	0	0	-4	-2	0	0	0.707	0.091
C6	2	2	-4	-2	0	0	0.891	0.114

Table 23: Environmentalist viewpoint - REMBRANDT Comparison Matrix for the Six Criteria and Their Resulting Weights

	C1	C2	C3	C4	C5	C6	Geo Mean	Weights
C1	0	-2	2	2	-4	2	1.000	0.147
C2	2	0	4	2	-2	2	1.587	0.233
C3	-2	-4	0	0	-2	0	0.630	0.093
C4	-2	-2	0	0	-2	0	0.707	0.104
C5	4	2	2	2	0	4	2.245	0.330
C6	-2	-2	0	0	-4	0	0.630	0.093

Table 24: Labor viewpoint - REMBRANDT Comparison Matrix for the Six Criteria and Their Resulting Weights

	C1	C2	C3	C4	C5	C6	Geo Mean	Weights
C1	0	-6	4	4	-2	2	1.122	0.150
C2	6	0	6	4	2	2	3.175	0.424
C3	-4	-6	0	0	-2	0	0.500	0.067
C4	-4	-4	0	0	-2	0	0.561	0.075
C5	2	-2	2	2	0	2	1.414	0.189
C6	-2	-2	0	0	-2	0	0.707	0.095

Table 25: Corporations viewpoint - REMBRANDT Comparison Matrix for the Six Criteria and Their Resulting Weights

Table 26: Concordance Matrix (Values Rounded to 3 Decimal Places)

SC	**SC1**	**SC3**	**SC9**	**SC10**	**SC11**	**SC12**	**SC13**	**SC16**	**SC18**	**SC20**
SC1	0.000	0.697	0.636	0.697	0.697	0.697	0.697	0.697	0.455	0.697
SC3	0.424	0.000	0.485	0.485	0.879	0.758	0.697	0.939	0.424	0.303
SC9	0.364	0.576	0.000	0.697	0.697	0.636	0.576	0.758	0.364	0.424
SC10	0.545	0.697	0.758	0.000	0.697	0.758	0.576	0.758	0.303	0.545
SC11	0.303	0.636	0.485	0.485	0.000	0.515	0.576	0.697	0.303	0.303
SC12	0.364	0.636	0.485	0.485	0.879	0.000	0.758	1.000	0.364	0.485
SC13	0.424	0.485	0.485	0.485	0.727	0.424	0.000	0.424	0.424	0.485
SC16	0.364	0.515	0.485	0.364	0.879	0.697	0.576	0.000	0.364	0.182
SC18	1.000	0.697	1.000	0.879	0.697	0.697	0.697	0.818	0.000	0.697
SC20	0.818	0.879	0.758	1.000	0.879	0.939	0.758	0.939	0.576	0.000

Table 27: Discordance Matrix

SC	**SC1**	**SC3**	**SC9**	**SC10**	**SC11**	**SC12**	**SC13**	**SC16**	**SC18**	**SC20**
SC1	0.0	0.0	0.0	0.0	1.0	1.0	1.0	0.0	0.0	1.0
SC3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
SC9	0.0	1.0	0.0	0.0	1.0	1.0	1.0	1.0	0.0	1.0
SC10	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
SC11	1.0	0.0	1.0	1.0	0.0	0.0	0.0	0.0	1.0	1.0
SC12	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0
SC13	1.0	0.0	1.0	1.0	0.0	0.0	0.0	0.0	1.0	1.0
SC16	0.0	0.0	1.0	0.0	0.0	0.0	1.0	0.0	1.0	0.0
SC18	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0
SC20	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

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